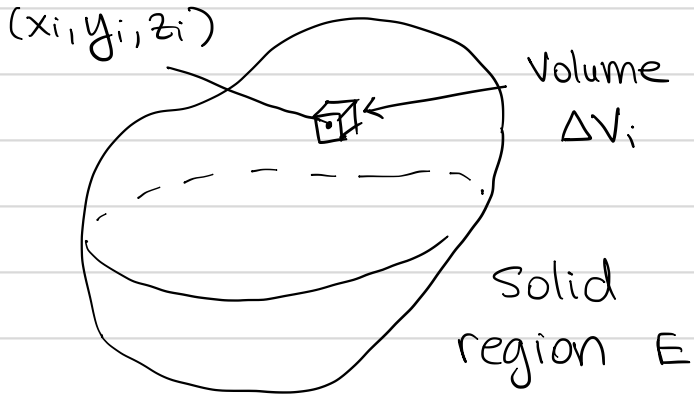
Triple integrals (15.6)



Definition:

$$\iiint_E f(x,y,z)\,dV = \lim_{\Delta V_i \to 0} \underbrace{\sum_i f(x_i,y_i,z_i)\,\Delta V_i}_{\text{Riemann sum}}$$

Applications

1) Volume (E) $= \iiint_E 1\,dV$

2) $\rho(x,y,z)$ = density (mass per unit volume) at (x,y,z)

$$\text{Mass}(E) = \iiint_E \rho(x,y,z)\,dV$$

3) Average of f in E $= \dfrac{1}{\text{Volume}} \iiint_E f\,dV$

Weighted avg. of f in E $= \dfrac{1}{\text{Mass}} \iiint_E f\rho\,dV$

4) Center of mass $= (\bar{x}, \bar{y}, \bar{z})$

$$\bar{x} = \frac{1}{\text{Mass}} \iiint_E x\rho\,dV$$

($\bar{y}$, $\bar{z}$ similar)

5) Moment of inertia

- about z-axis:

$$I_z = \iiint_E (x^2+y^2)\rho\,dV$$

(Rotational K.E.: $\frac{1}{2} I_z \omega^2$)

- about x-axis:

$$I_x = \iiint_E (y^2+z^2)\rho\,dV$$

- about y-axis:

$$I_y = \iiint_E (x^2+z^2)\rho\,dV$$

Fubini's theorem: still works!

Example: $B = [0,1]\times[-1,2]\times[0,3]$

$$f(x,y,z) = xyz^2$$

$$\iiint_B f\, dV = \int_0^1\int_{-1}^2\int_0^3 xyz^2 \underbrace{dz\,dy\,dx}_{\text{6 possible orders}}$$

$$= \int_0^1\int_{-1}^2 xy\left.\frac{z^3}{3}\right|_0^3 dy\,dx$$

$$= 9\int_0^1 x\left[\frac{y^2}{2}\right]_{y=-1}^{2} dx$$

$$= \frac{27}{2}\int_0^1 x\,dx = \frac{27}{4} \;//$$

$$\iiint_E f(x,y,z)\,dV =$$

$$= \iint_D \left(\int_{z_{min}(x,y)}^{z_{max}(x,y)} f\,dz\right) dA$$

$$= \int_{x_{min}}^{x_{max}}\int_{y_{min}(x)}^{y_{max}(x)}\int_{z_{min}(x,y)}^{z_{max}(x,y)} f\,dz\,dy\,dx$$

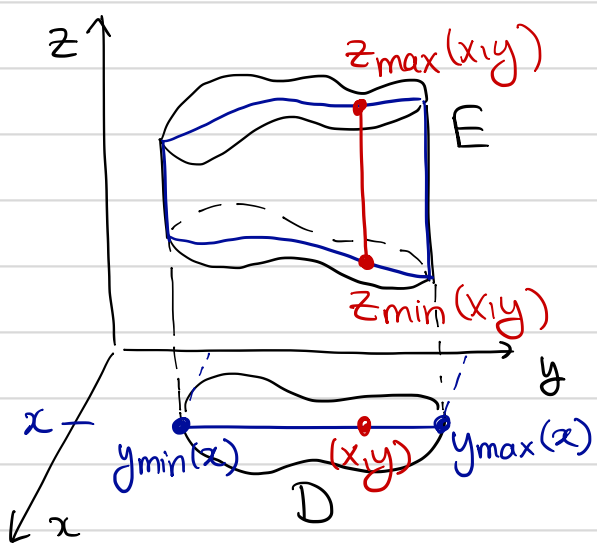
$$= \int_{x_{min}}^{x_{max}}\left(\iint_{x\text{-slice}} f\,dz\,dy\right) dx$$

E is a solid region of <u>type 1</u>

There exist also

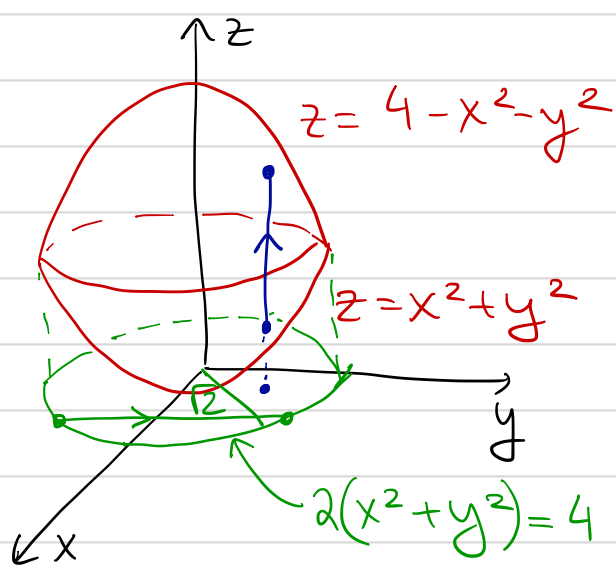
- type 2: D in yz-plane
- " 3: " " xz-plane

More general regions



Example: Volume of

$$E = \{(x,y,z) \mid x^2+y^2 \le z \le 4-x^2-y^2\}$$



$$\text{Vol} = \iiint_E 1 \cdot dV$$

$$= \int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{2-x^2}}^{\sqrt{2-x^2}} \int_{x^2+y^2}^{4-x^2-y^2} 1 \, dz \, dy \, dx$$

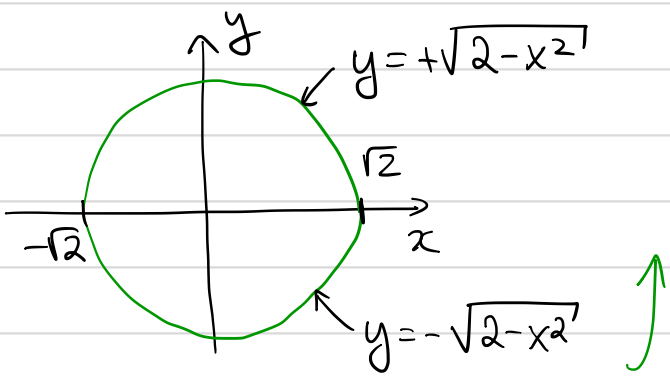
$$= \int_{\theta=0}^{2\pi} \int_{r=0}^{\sqrt{2}} \left(\int_{z=r^2}^{4-r^2} 1 \cdot dz \right) r \, dr \, d\theta$$

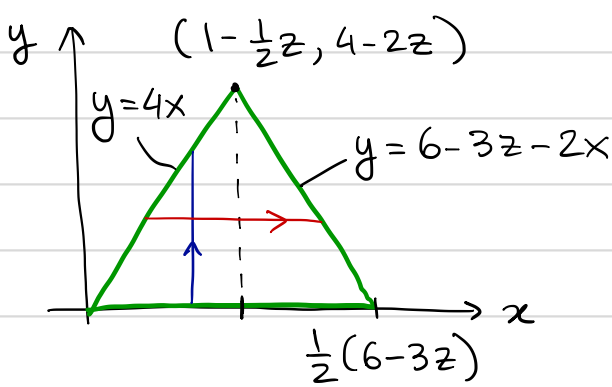
$$= \int_{\theta=0}^{2\pi} \left(\int_{r=0}^{\sqrt{2}} (4-r^2-r^2) \, r \, dr \right) d\theta$$

$$= 2\pi \cdot \int_0^{\sqrt{2}} 4r - 2r^3 \, dr$$

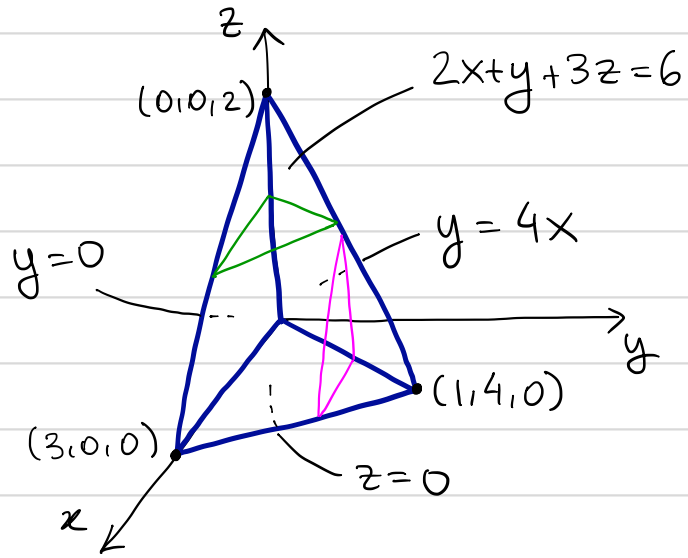
$$= 2\pi \left[4 \cdot \frac{r^2}{2} - 2 \cdot \frac{r^4}{4} \right]_{r=0}^{\sqrt{2}}$$

$$= 2\pi \left(2 \cdot 2 - \frac{1}{2} 4 \right) = \boxed{4\pi}$$





<u>Example</u>: E = tetrahedron



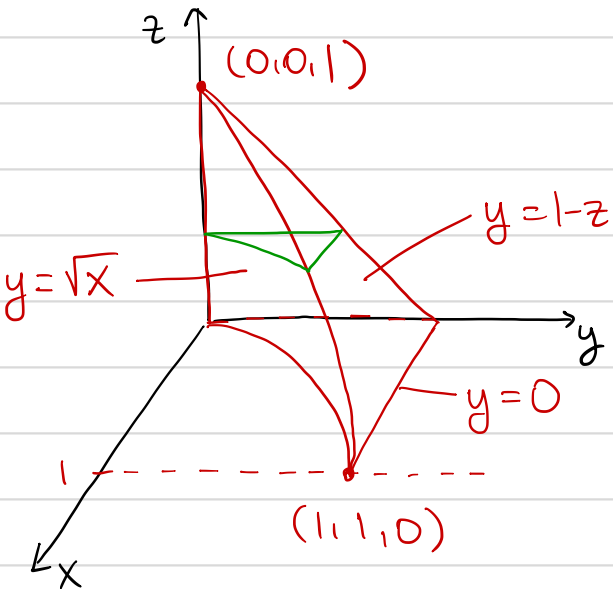
$$\iiint_E f(x,y,z) \, dV =$$

$$= \int_{z=0}^{2} \int_{y=0}^{4-2z} \int_{x=\frac{1}{4}y}^{\frac{1}{2}(6-y-3z)} f \, dx \, dy \, dz$$

$$= \int_{z=0}^{2} \left[\int_{x=0}^{1-\frac{1}{2}z} \int_{y=0}^{4x} f \, dy \, dx + \int_{x=1-\frac{1}{2}z}^{\frac{1}{2}(6-3z)} \int_{y=0}^{6-3z-2x} f \, dy \, dx \right] dz$$

$$= \int_{y=0}^{4} \int_{x=\frac{1}{4}y}^{\frac{1}{2}(6-y)} \int_{z=0}^{\frac{1}{3}(6-2x-y)} f \, dz \, dx \, dy$$

Example:

$$\int_{x=0}^{1}\int_{y=\sqrt{x}}^{1}\int_{z=0}^{1-y} f(x,y,z)\,dz\,dy\,dx =$$

$$= \int_{z=0}^{1}\int_{y=0}^{1-z}\int_{x=0}^{y^2} f(x,y,z)\,dx\,dy\,dz$$